
Extracorporeal Life Support in Patients with Congenital Diaphragmatic Hernia: How Long Should We Treat?

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- BACKGROUND:** Congenital diaphragmatic hernia (CDH) is a frequently lethal birth defect and, despite advances, extracorporeal life support (ie, extracorporeal membrane oxygenation [ECMO]) is commonly required for severely affected patients. Published data suggest that CDH survival after 2 weeks on ECMO is poor. Many centers limit duration of ECMO support.
- STUDY DESIGN:** We conducted a single-institution retrospective review of 19 years of CDH patients treated with ECMO, designed to evaluate which factors affect survival and duration of ECMO and define how long patients should be supported.
- RESULTS:** Of two hundred and forty consecutive CDH patients without lethal associated anomalies, 96 were treated with ECMO and 72 (75%) survived. Eighty required a single run of ECMO and 65 survived (81%), 16 required a second ECMO run and 7 survived (44%). Of patients still on ECMO at 2 weeks, 56% survived, at 3 weeks 46% survived, and at 4 weeks, 43% of patients still on ECMO survived to discharge. After 5 weeks of ECMO, survival had dropped to 15%, and after 40 days of ECMO support there were no survivors. Apgar score at 1 minute, Apgar score at 5 minutes, and Congenital Diaphragmatic Hernia Study Group predicted survival all correlated with survival on ECMO, need for second ECMO, and duration of ECMO. Lung-to-head ratio also correlated with duration of ECMO. All survivors were discharged breathing spontaneously with no support other than nasal cannula oxygen if needed.
- CONCLUSIONS:** In patients with severe CDH, improvement in pulmonary function sufficient to wean from ECMO can take 4 weeks or longer, and might require a second ECMO run. Pulmonary outcomes in these CDH patients can still be excellent, and the assignment of arbitrary ECMO treatment durations <4 weeks should be avoided. (J Am Coll Surg 2014;218:808–818. © 2014 by the American College of Surgeons)
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Congenital diaphragmatic hernia (CDH) is a severe birth defect with a broad spectrum of severity. Fortunately, many patients are born with low- and moderate-severity

defects and are cared for with standard ventilatory techniques. Patients with more severe CDH, however, often require extracorporeal life support (ie, extracorporeal membrane oxygenation [ECMO]) to maximize survival potential.^{1,2} Although gentle ventilatory techniques have proven to be a necessary component of successful therapeutic strategies,³ advances in ventilatory support, protocolized care, and wider use of nitric oxide and other pulmonary vasodilators have been unsuccessful in eliminating the need for ECMO in CDH patients.⁴⁻⁷

In CDH patients who require ECMO, survival rates are lower and complication rates are higher, which is intuitive, given that patients who require ECMO represent more severe disease than those who do not. Extracorporeal membrane oxygenation is associated with a myriad of additional risks, including bleeding, renal dysfunction, embolic phenomena, and compromised neurologic

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Abbreviations and Acronyms

CDH	= congenital diaphragmatic hernia
CDHSG	= Congenital Diaphragmatic Hernia Study Group
ECMO	= extracorporeal membrane oxygenation
LHR	= lung-to-head ratio
VA	= veno-arterial
VV	= veno-venous

outcomes.⁸⁻¹⁰ Based on published and center-specific data, it is not uncommon for centers to limit the total time of ECMO support in CDH patients who do not improve within assigned time frames. Tiruvoipati and colleagues found an 18% survival rate in CDH patients supported on ECMO >2 weeks¹¹ and, in discussions with other centers, we have found 2 weeks to be a common reference point for CDH survival on ECMO.

We were early adopters of the gentle ventilatory techniques taught by Wung and colleagues¹² and Boloker and colleagues,¹³ and helped define the negative effects of hyperventilation.¹⁴ These ventilation techniques form the foundation of our treatment strategy, and we have offered ECMO to patients who could not be adequately supported with these techniques, regardless of the severity of the underlying CDH physiology.

In this study, we review 19 consecutive years of CDH treatment at our institution to evaluate which patient factors affect survival and duration of ECMO, to evaluate the success of second ECMO runs when needed, to define the ECMO duration required for more severe CDH patients to recover, and to ask if there is a point at which additional ECMO support becomes futile.

METHODS

This is a retrospective review of consecutive patients with CDH treated at the University of Florida's Health and Shands Hospital for Children between September 1992 and December 31, 2011. A total of 268 CDH patients were identified from the cross reference of 2 separate medical record queries with operative records, autopsy records, a divisional database, and 2 prenatal evaluation databases. Patients with Morgagni CDH, diaphragmatic eventration, and patients in whom the diagnosis of CDH was missed and delayed more than 48 hours after delivery were not included. All patients were symptomatic in the first 6 hours of life. Of 268, twenty-eight (10%) were judged to have lethal associated anomalies.¹⁵ One hundred and five patients of the total were treated with ECMO, 9 of which were eventually judged to have associated lethal anomalies and are reported here for completeness and transparency, but are not analyzed

further. Ninety-six CDH patients, without lethal associated anomalies, were treated with ECMO and comprise the subjects of this study, which was approved by the University of Florida Institutional Review Board.

Clinical care

The majority of patients were prenatally diagnosed and counseled at our facility. Termination of CDH pregnancies did not occur in patients with nonlethal associated anomalies, regardless of severity, and, as such, terminations did not affect the survival results reported here. All patients were treated with strict limitation of ventilation pressures, avoidance of hyperventilation, and use of mild sedation as described previously.¹⁴ Medical oversight throughout the series was uniform, leading to a high degree of therapeutic consistency.

Extracorporeal membrane oxygenation was used for standard indications (oxygenation index > 40) and only when available modalities to avoid it (eg, pressors, nitric oxide, steroids, and milrinone) had failed. Either venovenous (VV) or veno-arterial (VA) ECMO was used, and patients who declined on VV ECMO were converted to VA. As the experience matured, VA ECMO was used preferentially for those patients judged by anatomy, lung-to-head ratio (LHR), and initial blood gas values to have more severe pulmonary hypoplasia.

Management on ECMO was not considered different from standard. Nitric oxide, if started before ECMO, was weaned off within 24 hours of initiating ECMO. Patients on high-frequency oscillatory ventilation were generally transitioned to conventional ventilation. Ultrafiltration and dialysis were strictly avoided. Representative resting ventilator settings were synchronized intermittent mandatory ventilation (SIMV) 30, inspiratory time of 0.6 seconds, PEEP of 6 cm H₂O, peak inspiratory pressure (PIP) of 22 cm of H₂O, and fraction of inspired O₂ of 0.4. Desired standards for discontinuing ECMO were satisfactory blood gases on SIMV ≤60 and fraction of inspired O₂ of ≤0.50. Patients successfully weaned from ECMO but who later declined and again met ECMO criteria were considered for a second run of ECMO. During the first 6 years of the study, patients were decannulated at ECMO cessation, and recannulated if a second ECMO run was necessary. Since then, we have perfused the ECMO cannulas with heparinized saline for 48 to 72 hours after cessation of ECMO before removing them, in case a second run of ECMO is needed. Second-run ECMO was also either VV or VA.

Patients were removed from ECMO when they met discontinuation criteria and when complications or additional organ failure made the likelihood of recovery appear dismal, but not for arbitrarily defined time end points.

Data collected and used for this analysis included gestational age, birth weight, Apgar scores at 1 (Apgar-1) and 5 minutes (Apgar-5), Congenital Diaphragmatic Hernia Study Group (CDHSG) predicted survival, side of defect, and liver position for left CDH. Use and type of ECMO, timing of ECMO and CDH repair, duration of ECMO, number of ECMO runs, and condition at discharge were also collected. Fifty-eight of 96 patients had prenatal lung-to-head ratio (LHR) measured and LHR observed-to-expected ratios calculated (<http://www.perinatology.com/calculators/LHR.htm>). Study data were collected and managed using REDCap electronic data capture tools hosted at the University of Florida.¹⁶

Analysis

Survival results were analyzed for the 96 patients in aggregate, and stratified by anatomic definitions of right, left liver down, and left liver up. Gestational age, birth weight, Apgar-1 and Apgar-5, and CDHSG predicted survival¹⁷ (<http://nicutools.org>) were tested for effects on survival. To evaluate duration of ECMO required to improve related to patient severity and anatomic variables, 4 patients who died early from ECMO complications (cannulation complications, oxygenator failure, and intracranial hemorrhage) were removed from the duration analysis only. Patients who required a second run of ECMO were also tested for associations with patient risk factors. Survivors only were then analyzed to define time required to improve on ECMO, and evaluated for correlations with patient severity factors, anatomical subsets, and CDHSG predicted survival.

Statistical analysis

We used the R statistical software package (V.3.0.2) to conduct all analyses and create all graphics. Logistic regression was used to model survival as a function of time on ECMO. Spearman's correlation or Mann-Whitney tests were used to evaluate the associations between ECMO duration and birth weight, Apgar-1 and Apgar-5, LHR, observed vs expected LHR, and type of initial ECMO (VV vs VA). We used *t*-tests or Mann-Whitney tests, as appropriate, to evaluate the associations between survival on ECMO and severity indices, and between the need for second ECMO and severity indices. ANOVA with the natural log of ECMO duration as the result, was used to evaluate the association between ECMO duration and anatomy. We used Fisher's exact test to evaluate the association between anatomy and survival and between anatomy and need for second ECMO. Finally, to facilitate visual comparison of expected ECMO durations by anatomy and by treatment sequence, we fit gamma distributions to the data using a best-fit algorithm (the "fitdistr" function in the R package MASS) and plotted these distributions side by side.

RESULTS

Of 268 consecutive CDH patients, 210 survived, for an overall survival of 78%. Of the 268 patients, 105 (39%) patients were supported with ECMO at some time before their discharge or death. The CDHSG predicted survival for these patients was 54 patients (51.5%) and 72 survived (69%, $p = 0.016$).

Twenty-eight of the 268 patients were judged to have lethal associated anomalies, and 210 of the remaining 240 (88%) survived. Of the 28 patients defined as having lethal associated anomalies, 9 received some support from ECMO, often only briefly, the full extent of the anomalies was assessed. These included 4 patients with bilateral CDH, 2 patients with severe combined vertebral and chest wall anomalies, 2 with cardiac defects (severe tricuspid valve anomaly and complete heart block), and 1 with hepatopulmonary fusion complicated by atrioventricular canal.

Of the 240 consecutive CDH patients without lethal associated anomalies, 96 (40%) patients were treated with ECMO at some point in their clinical course, representing the subjects of this study, and were analyzed further. Sixty-nine (72%) were inborn. Many included patients had associated anomalies that we considered nonlethal, and these included atrial septal defect ($n = 8$), ventricular septal defect ($n = 5$), coarctation of the aorta ($n = 1$), sacral dysgenesis requiring colostomy ($n = 1$), congenital cytomegalovirus infection ($n = 1$), and multiple patients with chromosomal anomalies, including inversion 13, 10q deletion, XXX 16q+, unbalanced translocation with partial trisomy 2, and partial trisomy 19 with a ring chromosome.

Seventy-two of 96 (75%) patients treated with ECMO survived to discharge. All patients were discharged home breathing spontaneously and on no support other than nasal cannula oxygen.

Surgical timing relative to extracorporeal membrane oxygenation

Table 1 reports the surgical timing related to ECMO and survival.

Proportion surviving vs time

Figure 1 shows the proportion of patients surviving vs days on ECMO for all 96 patients. Of patients who required >2 weeks of ECMO support, 56% survived; at 3 weeks, 46% survived; and at 4 weeks, 43% of patients still on ECMO survived to discharge. After 5 weeks of ECMO, survival had dropped to 15%, and after 40 total days of ECMO support there were no survivors. All survivors of ECMO runs longer than 3 weeks underwent CDH repair before 3 weeks.

Eighty of 96 patients (83%) received 1 ECMO run only and 65 survived (81%). A proportion surviving plot for

Table 1. Operative Status of Congenital Diaphragmatic Hernia Extracorporeal Membrane Oxygenation Patients

Operative status	n	Survived	
		n	%
Repaired before ECMO	30	27	90
Repaired during ECMO	34	23	68
Repaired after ECMO	25	22	88
Never repaired	7	0	0

ECMO, extracorporeal membrane oxygenation.

these patients (Fig. 2A) shows that, at 3 weeks, survival of patients still on ECMO was 62% and at 4 weeks, 100% of patients still on ECMO survived to discharge and 2 patients still on ECMO at 5 weeks survived.

Sixteen patients worsened after their first ECMO and were treated with a second run of ECMO (17%) and 7 survived to discharge (44%). Figure 2B shows the proportion surviving vs days on ECMO plot for second-run ECMO. At 2 weeks, survival was 40%; at 3 weeks, survival was 4 of 11 (31%); and by 4 weeks, survival had dropped to 1 of 10 (10%). After 31 days of total ECMO in patients who had 2 runs, there were no survivors out of 9 (0%).

Patient severity factors affecting survival on extracorporeal membrane oxygenation

Birth weight and gestational age did not correlate with survival on ECMO, but Apgar-1, Apgar-5, and CDHSG

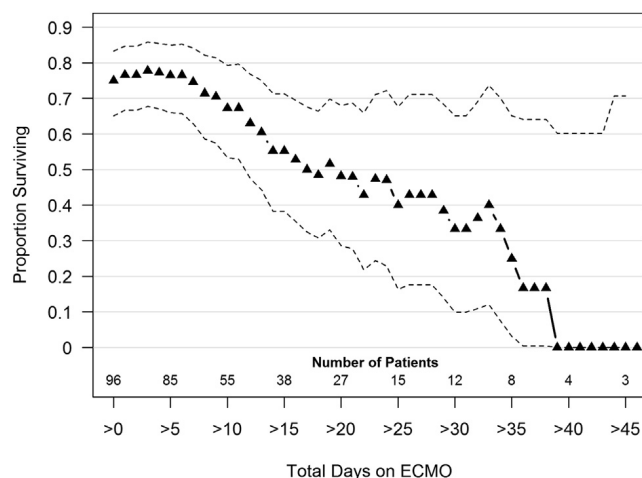


Figure 1. Proportion surviving vs days on extracorporeal membrane oxygenation (ECMO) curve for 96 consecutive congenital diaphragmatic hernia patients without lethal associated anomalies treated with ECMO. This plot includes both first- and second-run ECMO patients. Survival after 2 weeks on ECMO is 56%, after 3 weeks on ECMO is 46%, after 4 weeks on ECMO is 43%, and after 5 weeks on ECMO is 25%. Number of patients still on ECMO is shown along the bottom of the graph. Triangles represent the percent of patients still on ECMO at any time point who survive to discharge. Dotted lines represent confidence intervals.

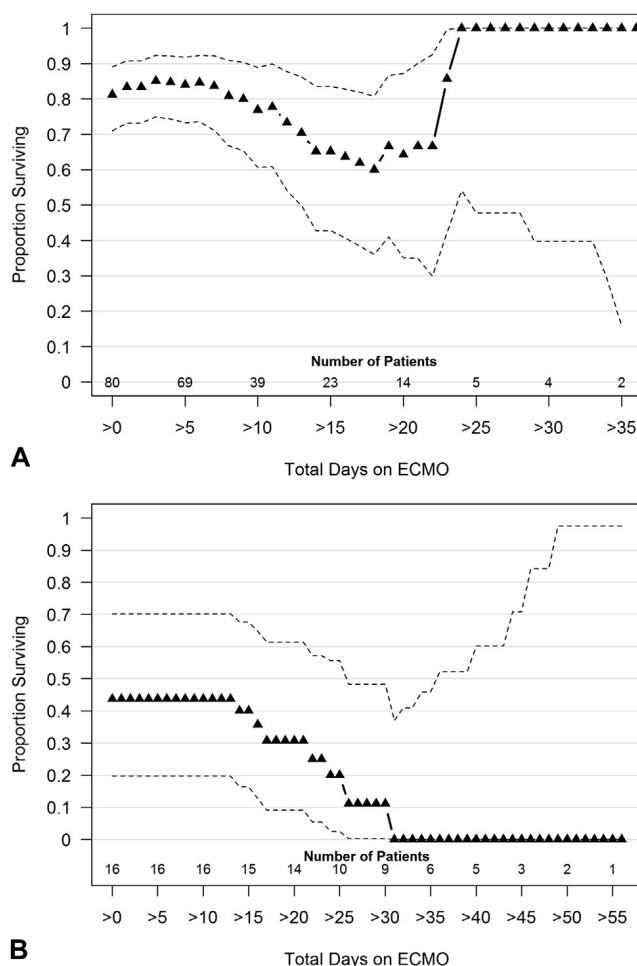


Figure 2. (A) Proportion surviving vs days on extracorporeal membrane oxygenation (ECMO) for 80 congenital diaphragmatic hernia patients who required just 1 ECMO run. (B) Proportion surviving vs days on ECMO for 16 patients treated with a second ECMO run. Number of patients still on ECMO is shown along the bottom of the graph. Triangles represent the percent of patients still on ECMO at any time point who survive to discharge. Dotted lines represent confidence intervals.

predicted survival were all strongly associated with survival on ECMO. Not all patients had an LHR measured (58 of 90 [60%]). The LHR and LHR observed-to-expected ratio were higher in the patients treated with ECMO who survived, but the difference was not statistically significant (Table 2).

Factors affecting need for a second run of extracorporeal membrane oxygenation

Gestational age and birth weight did not correlate with need for a second run of ECMO, Apgar-1 showed a marginal association, but Apgar-5 and CDHSG predicted survival showed a strong relationship with need for a second run of ECMO. Although the number of patients

Table 2. Correlation between Indices of Severity and Survival on Extracorporeal Membrane Oxygenation

	Total (n = 96)	Survivors (n = 72)	Nonsurvivors (n = 24)	p Value
Gestational age, mean $\pm$ SD	37.6 $\pm$ 1.8	37.7 $\pm$ 1.8	37.4 $\pm$ 1.9	0.563
Birth weight, mean $\pm$ SD	2,891 $\pm$ 475	2,903 $\pm$ 490	2,852 $\pm$ 435	0.635
Apgar at 1 min, mean $\pm$ SD (n = 95)	3.24 $\pm$ 2.1	3.61 $\pm$ 2.1	2.1 $\pm$ 1.5	0.002*
Apgar at 5 min, mean $\pm$ SD (n = 93)	5.70 $\pm$ 2.1	6.0 $\pm$ 2.1	4.7 $\pm$ 1.9	0.011*
LHR, mean $\pm$ SD (n = 58)	1.15 $\pm$ 0.5	1.20 $\pm$ 0.51	1.01 $\pm$ 0.48	0.210
LHR O/E, mean $\pm$ SD (n = 58)	29.4 $\pm$ 12.0	29.71 $\pm$ 11.1	28.4 $\pm$ 14.6	0.132
CDHSG predicted survival, n (%)	52 (54)	42 (58)	10 (42)	0.004*
Observed survival, n (%)	72 (75)			0.001*

*Significant.

CDHSG, Congenital Diaphragmatic Hernia Study Group; LHR, lung-to-head ratio; O/E, observed-to-expected ratio.

receiving prenatal LHR was a fraction of the total, LHR showed a marginal relationship with need for a second ECMO run (Table 3).

Duration of extracorporeal membrane oxygenation relative to severity markers

Duration of ECMO was not related to gestational age and birth weight, but was strongly related to Apgar-1, Apgar-5, CDHSG predicted survival, and LHR (Fig. 3). Linear regression defines the magnitude of effect that changes in each variable have on expected ECMO duration (Fig. 3).

Duration of extracorporeal membrane oxygenation relative to anatomy

Left liver-down CDH ECMO runs (n = 11, survivors = 10) were the shortest and had a mean length of 10.1 days, with a median of 9.5 days. Right CDH (n = 23, survivors = 19) and left liver-up CDH (n = 62, survivors = 43) runs were longer and similar in length, with a mean of 16.2 days and a median of 13.7 days compared with a mean of 17.1 days and a median of 12.2 days, respectively. This relationship did not reach statistical significance (overall ANOVA, p = 0.213).

Analysis of deaths on or after extracorporeal membrane oxygenation

Twenty-four patients treated with ECMO died before discharge, 15 after the first ECMO run and 9 after the

second. Based on the number of days until death and the cause of death, we categorized the timing of those deaths as early (related to early complications, generally <10 days), mid-course (complications related to repair on ECMO or failure to reach repair, generally $\leq$ 35 days of life), or late (after successful repair but failure to improve, generally >35 days) (Table 4).

First-run ECMO included 5 early deaths, 6 midcourse deaths, and 4 late deaths. Early deaths occurred from 2 technical complications and 2 intracranial hemorrhages. One patient never stabilized on VA ECMO and died early. Three patients failed to stabilize sufficiently on VV ECMO and, despite conversion to VA ECMO, died unrepaired around 3 weeks. Three patients who failed to wean from ECMO underwent repair on ECMO beyond 2 weeks and had complications of bleeding and abdominal compartment syndrome that truncated their ECMO courses. They died soon after. There were 4 late deaths due to failure of pulmonary hypertension to improve (n = 3) and fungal sepsis (n = 1).

Nine second-run ECMO deaths occurred. Three died mid-course related to abdominal compartment syndrome issues after repairs done after failing to wean from ECMO after 3 weeks. Three died due to late infections after prolonged courses on ECMO, and 3 died after long ECMO runs due to inability to improve pulmonary hypertension. These outcomes are summarized in Table 4.

Table 3. Relationship between Indices of Severity and Need for Second Extracorporeal Membrane Oxygenation

	Total, 1 run (n = 78), mean $\pm$ SD	Total, 2 runs (n = 16), mean $\pm$ SD	p Value
Birth weight	2,906 $\pm$ 485	2,814 $\pm$ 429	0.435 (NS)
Apgar at 1 min (n = 93)	3.4 $\pm$ 2.1	2.3 $\pm$ 1.7	0.061 (marginal)
Apgar at 5 min (n = 93)	5.9 $\pm$ 2.1	4.5 $\pm$ 2.1	0.034
CDHSG predicted survival (n = 93)	56.7 $\pm$ 21.2	40.7 $\pm$ 20.6	0.016
LHR (n = 58)	1.2 $\pm$ 0.5	0.90 $\pm$ 0.3	0.068 (marginal)
LHR O/E (n = 58)	30.6 $\pm$ 12.6	23.6 $\pm$ 6.9	0.106 (NS)

Patients with associated lethal anomalies are not included.

CDHSG, Congenital Diaphragmatic Hernia Study Group; LHR, lung-to-head ratio; O/E, observed-to-expected ratio.

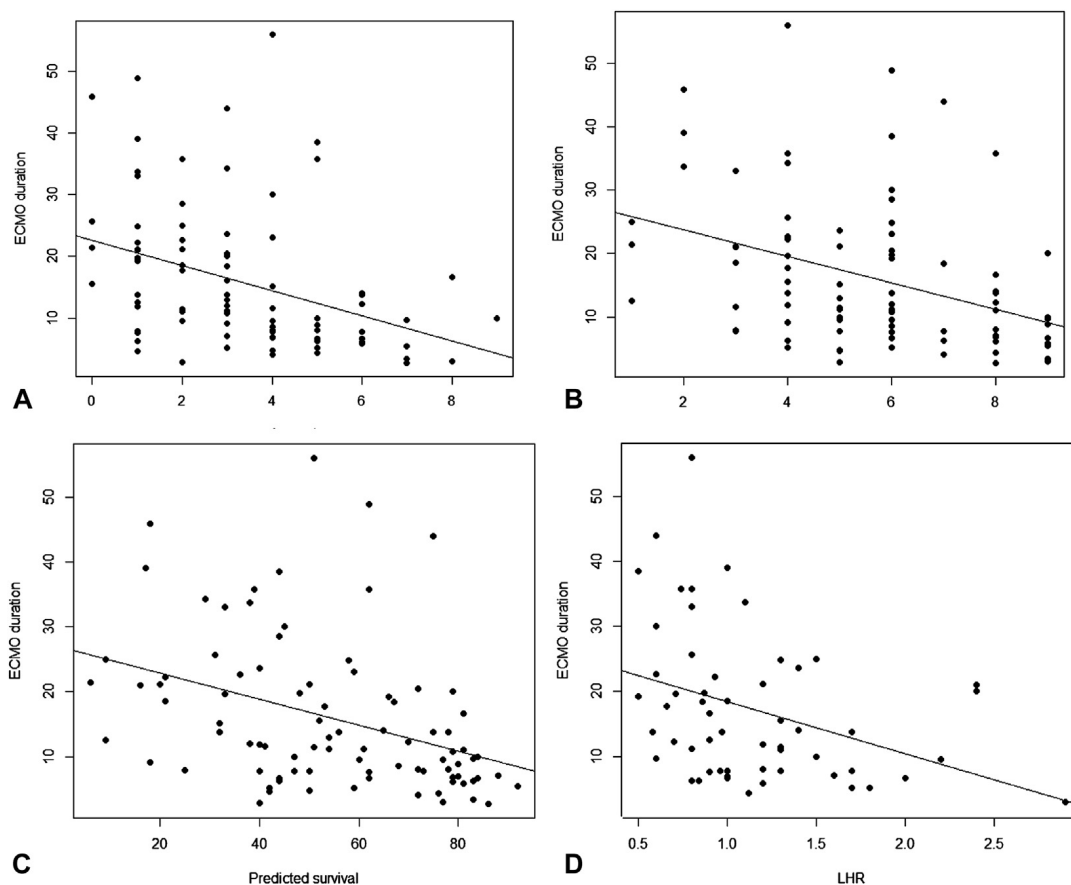


Figure 3. Relationship between severity markers and duration of extracorporeal membrane oxygenation (ECMO) run. (A) Apgar at 1 minute (Apgar-1). It is estimated that ECMO duration increases by 2.0 days for each unit decrease in Apgar-1 (95% CI, -3.11 to -0.964 ; $p < 0.001$). The plot below shows the data with a line of best fit. (B) Apgar at 5 minutes (Apgar-5). It is estimated that ECMO duration increases by 2.1 days for each unit decrease in Apgar-5 (95% CI, -3.16 to -1.00 ; $p < 0.001$). The plot below shows the data with a line of best fit. (C) Congenital Diaphragmatic Hernia Study Group predicted survival. It is estimated that ECMO duration increases by 2.0 days for each 10-unit decrease in survival score (95% CI, -0.306 to -0.097 ; $p < 0.001$). The plot below shows the data with a line of best fit. (D) Lung-to-head ratio (LHR). It is estimated that ECMO duration increases by 0.8 days for each 0.1 decrease in LHR (95% CI, -1.37 to -0.234 ; $p = 0.006$). The plot below shows the data with a line of best fit.

Survival related to initial type of extracorporeal membrane oxygenation

Forty-two of 56 patients who started on VV ECMO survived (75%) and 30 of 40 patients that started with VA ECMO survived (75%). Univariate analysis by chi-square shows no survival advantage related to initial ECMO type ($p = 1$). Multivariate regression analysis controlling for CDHSG predicted survival shows no survival advantage to initial ECMO type ($p = 0.467$).

Defining length of time to survive on extracorporeal membrane oxygenation (extracorporeal membrane oxygenation duration in survivors only)

To visually demonstrate the distribution of time on ECMO for survivors, probability density curves based

on the data from this series were created for each anatomic subset, for CDHSG predicted survival $>50\%$ and $<50\%$, and for LHR <1.0 or ≥ 1.0 (Fig. 4A–4C). Median length of ECMO duration to survival was shortest in left liver-down, intermediate in left liver-up, and longest in right CDH, but ANOVA was not statistically significant ($p = 0.44$). However, median ECMO duration to survival was significantly longer for patients with CDH predicted survival of $>50\%$ vs $\leq 50\%$ (13.1 days vs 9.5 days; $p = 0.006$) and for initial LHR <1.0 vs ≥ 1.0 (13.7 days vs 9.5 days; $p = 0.012$).

DISCUSSION

This report focuses on the ECMO duration and survival outcomes of 96 patients that were part of a large

Table 4. Causes of Death and Death Timing in Congenital Diaphragmatic Hernia Extracorporeal Membrane Oxygenation Patients

Cause	Total		First run		ECMO days,		Death DOL,		ECMO stopped,		Second run		ECMO days,		Death DOL,		ECMO stopped,	
	n	%	n	%	mean $\pm$ SD		mean $\pm$ SD		mean $\pm$ SD		n	%	mean $\pm$ SD		mean $\pm$ SD		mean $\pm$ SD	
Early (n = 5)																		
ECMO complication	2	8.3	2	13	0		5 $\pm$ 4		0		0							
ICH	2	8.3	2	13	5 $\pm$ 3		6 $\pm$ 4		0		0							
Never stabilized-VA	1	4.2	1	6.6	2.9		3		0		0							
Mid (n = 9)																		
Surgical-bleeding	2	8.3	2	13	16 $\pm$ 8		28 $\pm$ 19		11 $\pm$ 11		0							
Surgical-compartment	4	17	1	6.6	19		22		0		3	33	32 $\pm$ 5		35 $\pm$ 6		1 $\pm$ 1	
Never stabilized-VV	3	13	3	20	22 $\pm$ 2		23 $\pm$ 3		0		0							
Late (n = 10)																		
Sepsis	4	17	1	6.6	22		38		0		3	33	45 $\pm$ 5		58 $\pm$ 4		4 $\pm$ 4	
Pulmonary hypoplasia/HTN	6	25	3	20	15 $\pm$ 9		124 $\pm$ 70		107 $\pm$ 67		3	33	43 $\pm$ 12		59 $\pm$ 27		4 $\pm$ 4	
Total	24		15								9							

DOL, day of life; ECMO, extracorporeal membrane oxygenation; HTN, hypertension; ICH, intra-cranial hemorrhage; VA, veno-arterial; VV, veno-venous.

consecutive series of CDH. Although patients deemed to have associated lethal anomalies are excluded from this analysis, no patients were excluded due to severity alone, and ECMO was used inclusively in an attempt to rescue patients that were not known to have other anomalies that would preclude survival. That 72% were inborn, 89% were either left liver-up or right CDH, and the aggregate CDHSG predicted survival was 54%, all attest to the overall clinical severity of this series.

This study demonstrates several important findings. First, severity markers, including Apgar-1, Apgar-5, and CDHSG predicted survival, correlated strongly with survival on ECMO, need for second ECMO run, and duration of ECMO. The CDHSG predicted survival correlated the most strongly of the severity markers. Seetharamaiah and colleagues previously showed that Apgar-5 correlated with survival in CDH patients treated with ECMO but, in our series, we found no outcomes associations with gestational age or birth weight.¹⁸ Although the correlations between known markers of severity and ECMO outcomes might seem intuitive, their clear demonstration allows the clinician to paint an increasingly granular picture of CDH severity and to better understand how that severity will affect the ECMO run.

The LHR measurements were not available in all patients, which hampered the ability to test associations with outcomes measures. However, LHR values varied inversely and strongly with duration of ECMO, clarifying that native lung mass influences time on ECMO. This is important, suggesting that ECMO for CDH patients not only allows time for reactive pulmonary hypertension to resolve, a process that would be expected to be similar in duration in patients regardless of severity, but that additional time is required for smaller lungs to accommodate. From these data, we can only postulate what that accommodation consists of, but suggest that multiple processes with a combined longer time course than simply the resolution of reactive pulmonary hypertension are operative. It is tempting to suggest that some degree of lung growth might be occurring, but this was not directly tested here.

In patients with LHR <1.0, representing severe pulmonary hypoplasia, survival to discharge was achieved in 17 of 26 (65%) patients, significantly higher than many reports would suggest.¹⁹⁻²¹ Some might postulate that the LHR measurements were inaccurate but, given the large size and inclusive nature of this series, the distribution of patients with LHR <1.0 is consistent with other series.²² The survival of two thirds of these patients is likely related to treatment variables, of which duration of ECMO, as reported here, and repair of liver-up CDH before ECMO are treatment factors, which might have

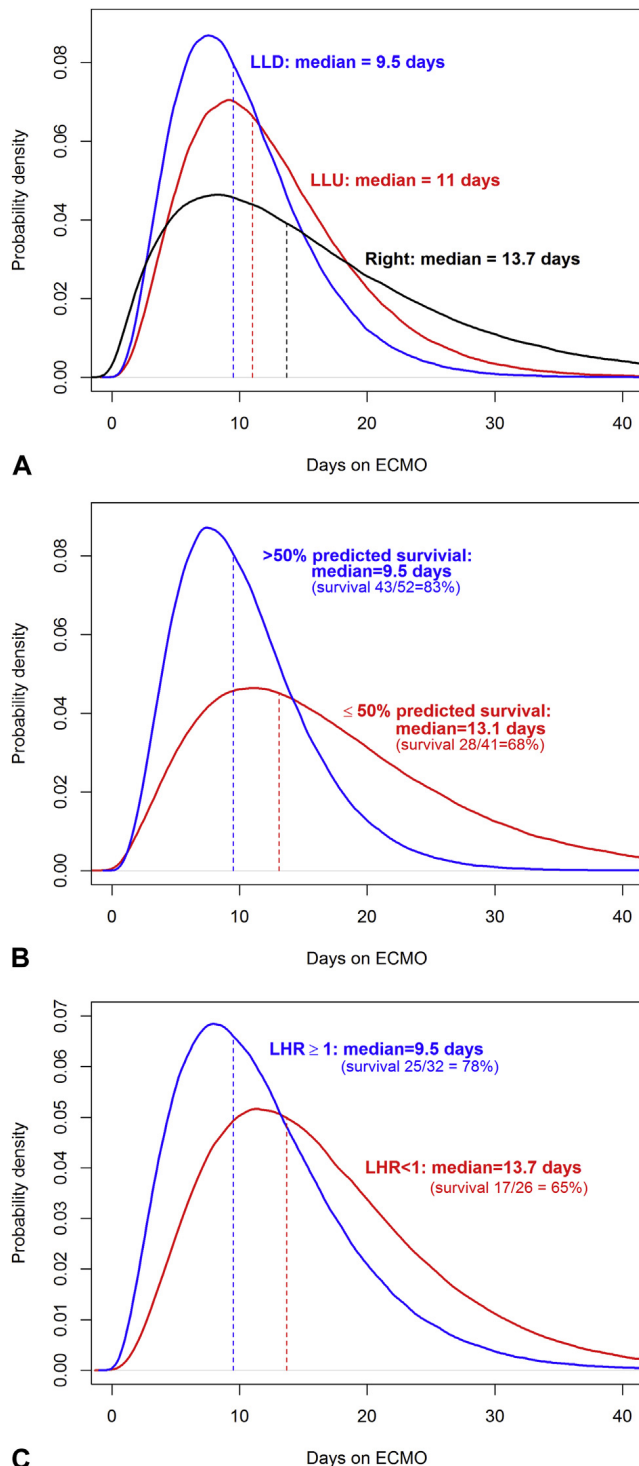


Figure 4. Comparison of extracorporeal membrane oxygenation (ECMO) duration in survivors based on anatomy, Congenital Diaphragmatic Hernia Study Group (CDHSG) predicted survival, and lung-to-head ratio (LHR). (A) Distribution curve of estimate ECMO duration for patients with congenital diaphragmatic hernia (CDH) anatomy of left liver-down (LLD) vs left liver-up (LLU) vs right CDH ($p = 0.448$). (B) Predicted survival. Distribution curve of estimated ECMO duration in patients with CDHSG predicted survival $>50\%$

contributed to the survival obtained in these severely affected patients. Optimal lung adaptation and accommodation in patients with severe pulmonary hypoplasia are likely facilitated by repair, and repair before ECMO in liver-up CDH was associated with improved survival in our recent report.¹⁵

These data show that patients with more severe CDH, as defined by a variety of patient-specific and reproducible markers, have higher mortality and higher risk for second-run ECMO, but also need a longer duration of ECMO to survive. To optimize survival in these most severe patients, centers must be willing to invest considerable time in their recovery on ECMO. However, it is not as simple as just investing more time because the more severe CDH patients are less tolerant of complications and missteps, as noted in our own analysis of deaths. It is not an accident that the severity markers correlate so strongly with risk of death and risk of second ECMO, and this clarifies the necessity of exacting care to maximize the recovery potential in these infants.

This concept applies to the patient who is weaning from ECMO as well. Although the numbers in the series get smaller as the days on ECMO increase, the message of the first- and second-run ECMO survival curves suggests it is likely better to invest enough time to come off ECMO in optimal condition versus not. Controlling for number of days on ECMO, survival with first-run ECMO was always better than second-run ECMO. This can help inform decision making when a choice arises between investing more time on ECMO vs coming off under less than optimal circumstances due to ECMO-specific issues. The less-severe patient, as defined by risk markers, can tolerate discontinuing ECMO early, but the more-severe patient likely will not. Based on these data, we infer the importance of optimization of clinical status before discontinuing ECMO whenever possible, to minimize the need for a second ECMO run and an associated decreased chance of survival.

Similarly, we interpret these data to show the reason for patience as patients wean from ECMO. Efforts to get patients off ECMO sooner by escalating ventilation strategies might ultimately compromise the patient's chance of recovery, as higher ventilator settings can increase the long-term risk of unresolved pulmonary hypertension.²³

To better understand the issues about CDH survival and duration on ECMO, we believed it was important to define when and why patients died. We learned that 10 (42%) of 24 patients died of late issues of infection

compared with $\leq 50\%$ ($p = 0.006$). (C) LHR. Distribution curve of estimated ECMO duration for patients with LHR ≥ 1 vs < 1 ($p = 0.012$). All curves based on data in this series.

and unresolved pulmonary hypertension, but that nearly 60% died of early and mid-course issues that are potentially remediable. Although the severity markers show that those who die in mid-course are much sicker than those who do not, these patients went on ECMO unrepaired and failed to improve for ≥ 2 weeks, resulting in decisions to repair on ECMO. Techniques such as early repair on ECMO⁸ or repair before ECMO in patients who have such an opportunity,¹⁵ might provide better options for salvaging these specific high-risk patients. The unsolved challenge is defining who these patients are early in the treatment course so that severity-specific treatment algorithms can be developed and applied.^{19,24}

We interpret the totality of the data presented here to show that although many patients will be weanable from ECMO in 1 to 2 weeks, those patients with more severe CDH will simply take longer, and discontinuation of ECMO based solely on arbitrary times on ECMO should be re-evaluated where practiced. In patients who die, the majority will die from complications that are obvious and the need to discontinue ECMO will be clear.

The major weakness of this study is that, despite its size, the number of patients still on ECMO at 1 month is small, limiting the strength of observation at that point. Second, this study suffers from the weaknesses of a retrospective review, but as a single-institution series with consistent therapeutic leadership, the series has considerable strengths as well. The ability to analyze deaths in a granular manner is a strength not applicable to multicenter databases. Third, the survival attained in this series is high, and the outcomes related to specific durations of ECMO might not be directly applicable to all centers, although the general points should be. Fourth, some might question if using ECMO in 40% of CDH patients is representative, but our approach is inclusive, and this use is in line with other contemporary reports,^{1,4,8} and this rate has not changed from our previous report.¹⁴ Finally, survival in CDH is only part of the story and quality of outcomes is an essential piece. Many authors have documented the neurodevelopmental challenges faced by this group of patients,¹⁷⁻²⁰ and we are currently evaluating the imaging and neurodevelopmental outcomes of our patients for subsequent reporting.

CONCLUSIONS

Despite advances in the care of CDH patients, ECMO support remains an important tool in the care of many of these infants. Patients who are more severe, as measured by a variety of standard markers, are at higher risk of dying and also require more time on ECMO to recover. Survival is maximized by optimizing first-run

ECMO. Patients who worsen gain after a first run of ECMO should be offered a second run unless complications intervene. At our center, survival at 4 weeks of ECMO is 43% and at 5 weeks is 25%, but we had no survivors of second-run ECMO past 30 days and no survivors of first-run ECMO past 40 days. All patients treated with >3 weeks of ECMO and ultimately surviving were repaired before that time. The reasons for death on ECMO are usually visible and definable, and we caution against defining arbitrary lengths of ECMO shorter than 4 weeks in stable patients who have been successfully repaired, especially in those more severely affected patients who, as described here, need more time for lung adaptation and accommodation.

Author Contributions

Study conception and design: Kays, Islam, Larson, Perkins, Talbert

Acquisition of data: Kays, Islam, Richards, Larson, Perkins

Analysis and interpretation of data: Kays, Islam, Richards, Larson, Perkins, Talbert

Drafting of manuscript: Kays, Islam, Richards, Larson, Perkins, Talbert

Critical revision: Kays, Islam, Richards, Larson, Talbert

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Discussion

DR EUGENE MCGAHREN (Charlottesville, VA): Dr Kays addresses a population of patients, babies with congenital diaphragmatic hernia of the Bochdalek type, which has historically been very difficult to study. The course of each individual baby can be so unpredictable that it has been difficult to gain new understanding of their care in a prospective manner. However, over time, and primarily in retrospective fashion, we have learned some important principles.

Diaphragmatic hernia is a medical emergency, not a surgical one. Optimal ventilatory support strategies use the concept of gentle ventilation, also known as permissive hypercapnia; a strategy that Dr Kay's group adopted and published soon after its introduction at Columbia, and that our group presented on at this meeting 2 years ago.

We have also learned that extracorporeal membrane oxygenation (ECMO) is a valuable salvage modality. In retrospective fashion, Dr Kays aimed to provide more information as to how ECMO is best used in this population of babies. In particular, he and his group indicated that long initial ECMO courses, and even second ECMO courses, may allow survival in the sickest babies with the worst predictive parameters. This would be counter to other reports that suggest that ECMO should not be offered at all to some of these babies, or at least should be limited in its time course if it is indeed offered at all. I have a few questions for Dr Kays.

First, how was the decision made for timing of operative repair relative to the ECMO course, and what recommendations would you have regarding that?

Second, were there more anomalies in the babies in your series with ECMO courses than those without ECMO? And particularly, were there more babies with anomalies in the longer courses?

Third, do we have any survival data beyond discharge for babies who required ECMO, and, again, particularly in the babies with the long courses?

Fourth, we do know that neurodevelopmental delay can be present in babies who have congenital diaphragmatic hernia and require ECMO. What are the plans for studying these babies as you move forward, particularly the babies with the long courses?

Finally, how do we apply what we have learned today to our practice? In other words, how do we know when to stop an ECMO course? There are some courses in which the endpoint is self-evident, perhaps by bleeding or compartment syndrome. But there are some babies you describe in the manuscript in whom the course was stopped because of failure to wean. So when do we actually get to the point of understanding when we have failure to wean?

Congratulations to Dr Kays and his group for a paper that increases our fund of knowledge of the management of this very